

EC4219: Software Engineering

Lecture 11 — Program Verification (5) *Termination Proof*

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Total Correctness

- Total correctness of a function asserts that, if the precondition holds at entry, then the function eventually halts and the postcondition holds.
- Total correctness = Partial correctness + **Termination**

Well-Founded Relations

Termination proof is based on well-founded relations.

- A binary relation $\prec$ (“precedes” or “is less than”) over a set S is **well-founded** iff there does not exist an infinite decreasing sequence $s_1, s_2, \dots$ of elements of S :

$$s_1 \succ s_2 \dots$$

where $s \prec t$ iff $t \succ s$. Equivalently, $\prec$ is well-founded iff every non-empty subset $X \subseteq S$ has a minimal element under $\prec$.

- For example, the relation $<$ is well-founded over the natural numbers (including 0), because any sequence of natural numbers that decrease according to $<$ is finite, e.g., $1023 > 39 > 30 > 29 > 8 > 0$.
- However, $<$ is not well-founded over the rationals $\mathbb{Q}$, e.g., $X = \{\frac{1}{i} \mid i \in [1, n]\} \subseteq \mathbb{Q}$ has an infinitely decreasing sequence $1 > \frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \dots$.

Lexicographic Relations

- A useful class of well-founded relations.
- From a set of pairs of sets and well-founded relations:

$$(S_1, \prec_1), \dots, (S_m, \prec_m)$$

construct the set

$$S = S_1 \times \dots \times S_m$$

and define the relation $\prec$:

$$(s_1, \dots, s_m) \prec (t_1, \dots, t_m) \iff \bigvee_{i=1}^m (s_i \prec_i t_i \wedge \bigwedge_{j=1}^{i-1} s_j = t_j)$$

where $s_i, t_i \in S_i$.

- For example, let $S = \mathbb{N}^3$ be the set of triples of natural numbers, and $<_3$ be the lexicographic extension of $<$ to triples. Then,

$$(11, 9, 104) <_3 (11, 13, 3)$$

Proving Termination

- Define a set S with a well-founded relation $\prec$.
 - ▶ We usually choose the set of n -tuples of natural numbers as S , and choose $<_n$ (the lexicographic extension of $<$) as $\prec$.
 - ▶ For example, if $n = 2$,

$$(a, b) <_2 (a', b') \iff a < a' \vee (a = a' \wedge b < b')$$

- Find a **ranking function** δ that maps program states (at each iteration) to S such that δ decreases according to $\prec$ along every basic path.
 - ▶ Find a **decreasing expression** (note: this is not a standard term!)
- Then, since $\prec$ is well-founded, there cannot exist an infinite sequence of program states.

In summary, our goal is to show that there are some decreasing expressions whose lower bounds are 0.

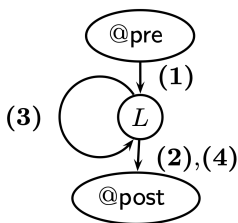
Example 1: Linear Search

```
1  @pre:  $0 \leq l \wedge u < |a|$ 
2  @post:  $rv \leftrightarrow \exists i. l \leq i \leq u \wedge a[i] = e$ 
3  bool LinearSearch (int  $a[]$ , int  $l$ , int  $u$ , int  $e$ ) {
4      int  $i := l$ ;
5      while ( $i \leq u$ ) {
6          @L:  $u < |a|$ 
7           $\downarrow (|a| - i)$ 
8          if ( $a[i] = e$ ) return true;
9           $i := i + 1$ ;
10     }
11     return false;
12 }
```

Example 1: Linear Search

$@L : u < |a|$
 $\downarrow (|a| - i)$
assume $i \leq u$;
assume $a[i] \neq e$;
 $i := i + 1$;
 $\downarrow (|a| - i)$

The other basic paths ((1), (2), (4)) are not relevant to proving termination.



Example 2: Bubble Sort

```
1  @pre:  $\top$ 
2  @post:  $\top$ 
3  bool BubbleSort (int  $a_0[]$ ) {
4      int[]  $a := a_0$ ;
5      @ $L_1 : i + 1 \geq 0$ 
6       $\downarrow (i + 1, i + 1)$ 
7      for (int  $i := |a| - 1$ ;  $i > 0$ ;  $i := i - 1$ ) {
8          @ $L_2 : i + 1 \geq 0 \wedge i - j \geq 0$ 
9           $\downarrow (i + 1, i - j)$ 
10         for (int  $j := 0$ ;  $j < i$ ;  $j := j + 1$ ) {
11             if ( $a[j] > a[j + 1]$ ) {
12                 int  $t := a[j]$ ;
13                  $a[j] := a[j + 1]$ ;
14                  $a[j + 1] := t$ ;
15             }
16         }
17     }
18     return  $a$ ;
19 }
```


Example 2: Bubble Sort

(1)

$@L_1 : i + 1 \geq 0$

$\downarrow (i + 1, i + 1)$

assume $i > 0$;

$j := 0$;

$\downarrow (i + 1, i - j)$

(2) true branch of the inner loop

$@L_2 : i + 1 \geq 0 \wedge i - j \geq 0$

$\downarrow (i + 1, i - j)$

assume $j < i$;

assume $a[j] > a[j + 1]$;

$t := a[j]$;

$a[j] := a[j + 1]$;

$a[j + 1] := t$;

$j := j + 1$;

$\downarrow (i + 1, i - j)$

Example 2: Bubble Sort

(3) false branch of the inner loop

$@L_2 : i + 1 \geq 0 \wedge i - j \geq 0$

$\downarrow (i + 1, i - j)$

assume $j < i$;

assume $a[j] \leq a[j + 1]$;

$j := j + 1$;

$\downarrow (i + 1, i - j)$

(4)

$@L_2 : i + 1 \geq 0 \wedge i - j \geq 0$

$\downarrow (i + 1, i - j)$

assume $j \geq i$;

$i := i - 1$;

$\downarrow (i + 1, i + 1)$

The other basic paths are not relevant to proving termination.

Verification Conditions

$$\begin{array}{l} @F \\ \downarrow \delta[\bar{x}] \\ S_1; \\ \vdots; \\ S_n; \\ \downarrow \kappa[\bar{x}] \end{array}$$

The verification condition of the basic path is:

$$F \rightarrow \mathbf{wp}(\kappa \prec \delta[\bar{x}_0], S_1; \cdots S_n) \{ \bar{x}_0 \mapsto \bar{x} \}$$

Our hope is to show that κ after executing the statements is less than δ before executing the statements.

Example

Derive the VC for the basic path:

(4)

$@L_2 : i + 1 \geq 0 \wedge i - j \geq 0$

$\downarrow (i + 1, i - j)$

assume $j \geq i$;

$i := i - 1$;

$\downarrow (i + 1, i + 1)$

Example

Derive the VC for the basic path:

(4)

$@L_2 : i + 1 \geq 0 \wedge i - j \geq 0$

$\downarrow (i + 1, i - j)$

assume $j \geq i$;

$i := i - 1$;

$\downarrow (i + 1, i + 1)$

First, compute the weakest precondition:

$$\begin{aligned} & \mathbf{wp}((i + 1, i + 1) <_2 (i_0 + 1, i_0 - j_0), \mathbf{assume } j \geq i; i := i - 1) \\ & \iff \mathbf{wp}((i - 1) + 1, (i - 1) + 1) <_2 (i_0 + 1, i_0 - j_0), \mathbf{assume } j \geq i) \\ & \iff j \geq i \rightarrow (i, i) <_2 (i_0 + 1, i_0 - j_0) \end{aligned}$$

Next, rename the variables:

$$j \geq i \rightarrow (i, i) <_2 (i + 1, i - j)$$

The final VC:

$$L_2 \rightarrow j \geq i \rightarrow (i, i) <_2 (i + 1, i - j)$$

Summary

Termination proof:

- Choose the set of n -tuples of natural numbers as S , and choose $<_n$ as $\prec$.
- Find a ranking function (expression) that decreases over S under $<_n$.
- If the ranking function decreases along every basic path, we prove termination.

Next class: type system